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02

Data replication and migration

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Data replication and migration

01 Replication and migration architecture

02 The `gcloud` command line tool

03 Moving datasets

04 Datastream

Google Cloud

In this module, first, you review the baseline Google Cloud data replication and migration architecture.

Second, you will cover the options and use cases for the gcloud command line tool.

Then you will review the functionality and use cases for the Storage Transfer Service.

The next topic addresses the functionality and use cases for the Transfer Appliance.

Finally, you will look at the features and deployment of Datastream.

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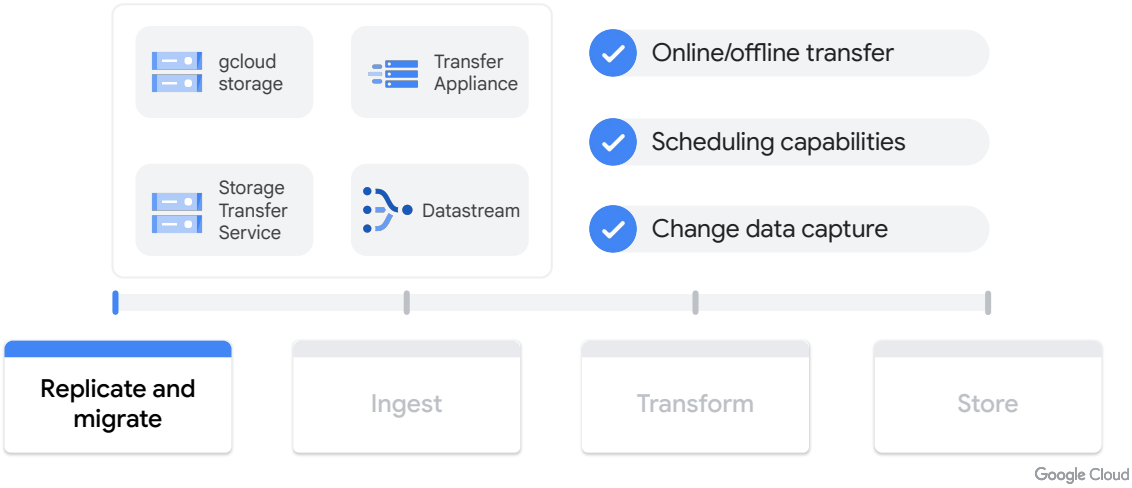
Replication and migration services onboard your data into Google Cloud



Google Cloud

The replicate and migrate stage of a data pipeline focuses on the tools and options to bring data from external or internal systems into Google Cloud for further refinement.

Google Cloud provides a comprehensive suite of tools to migrate and replicate your data.

Start replicating and migrating data by using tools like the “gcloud storage” command, Transfer Appliance, Storage Transfer Service, or Datastream.

You can then transform the data as needed before finally storing it within Google Cloud.

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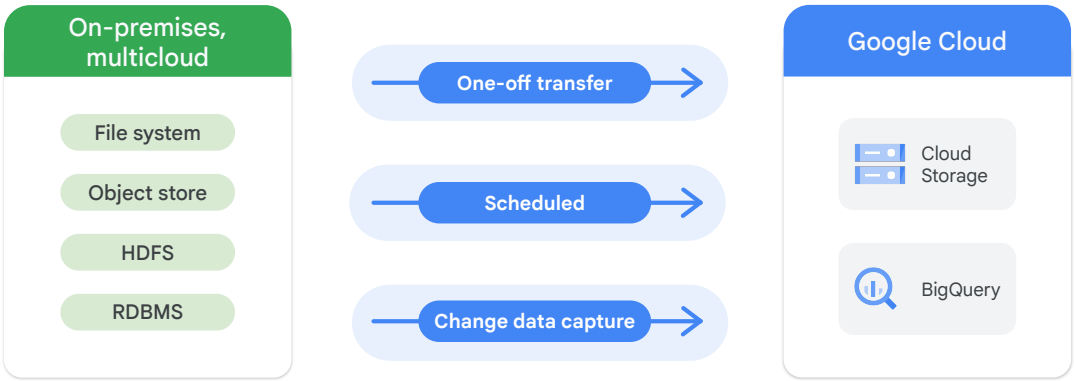
Common replication and migration scenarios for ingesting your data into Google Cloud



Google Cloud

Data can originate from on-premises or multicloud environments, including file systems, object stores, HDFS, and Relational Databases.

Google Cloud offers options for one-off transfers, scheduled replications, and change data capture, ultimately landing data in Cloud Storage or BigQuery.

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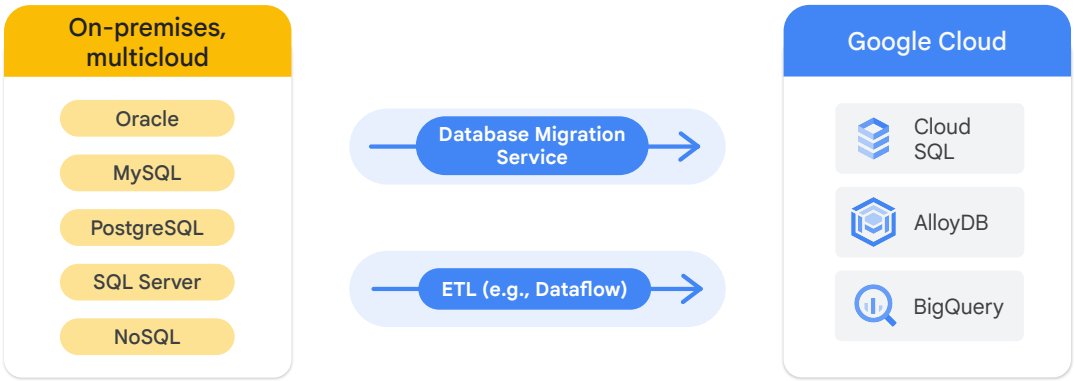
Google Cloud provides additional options for migrating your workloads



Google Cloud

Google Cloud provides additional workload migration with options for various database types.

Leverage Database Migration Service for seamless transitions from Oracle, MySQL, PostgreSQL, and SQL Server. For other data formats or complex migrations, use ETL tools like Dataflow with a wide range of templates that handle NoSQL or non-relational databases.

Your target destination can be Cloud SQL, AlloyDB, or BigQuery depending on your needs.

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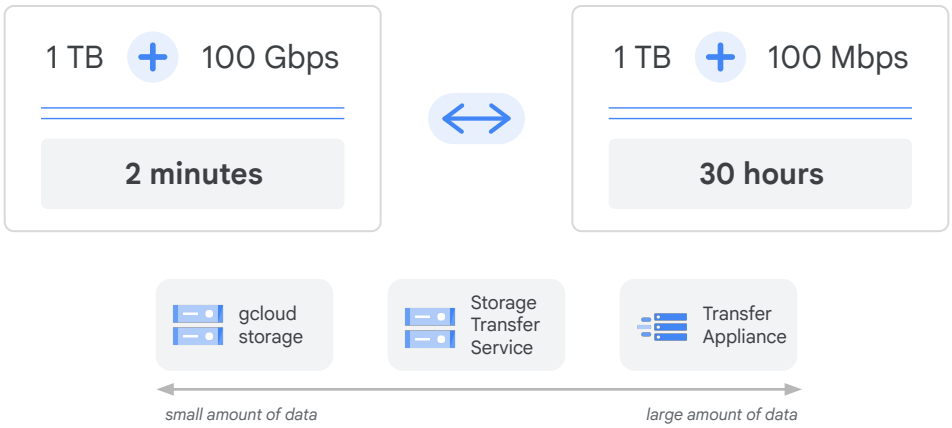
Choosing between migration options depends on your amount of data and network bandwidth



Google Cloud

The ease of migrating data depends heavily on data size and network bandwidth.

With 1 Terabyte of data, a 100 Gigabits per second network takes about 2 minutes to transfer, while the same size on a 100 Megabits per second network takes 30 hours.

The "gcloud storage" command or Storage Transfer Service are suitable for smaller datasets. For larger datasets, consider Transfer Appliance for faster offline transfer.

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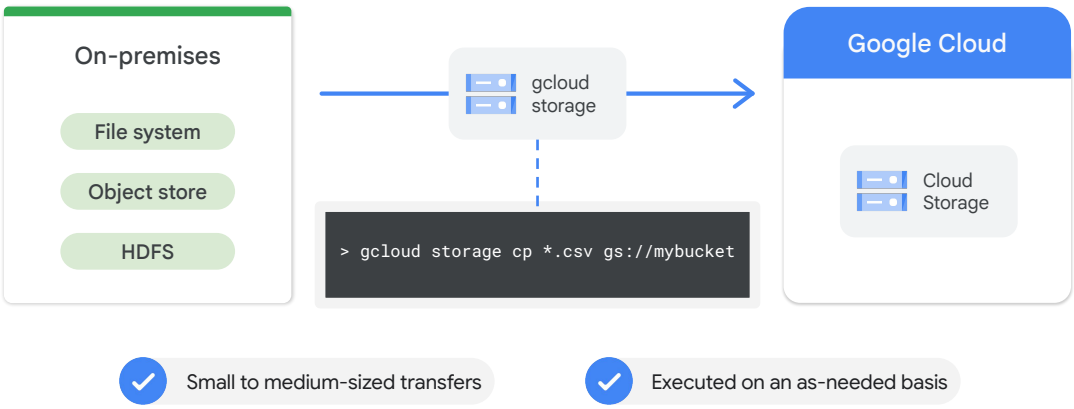
Use the `gcloud storage` command to move small amounts of data to Cloud Storage



- Small to medium-sized transfers
- Executed on an as-needed basis

Google Cloud

You can use the “gcloud storage” command to transfer small to medium-sized datasets to Cloud Storage.

The data can originate from various on-premise sources like file systems, object stores, or HDFS.

The “cp” command, shown in the code snippet, facilitates these ad-hoc transfers directly to Cloud Storage.

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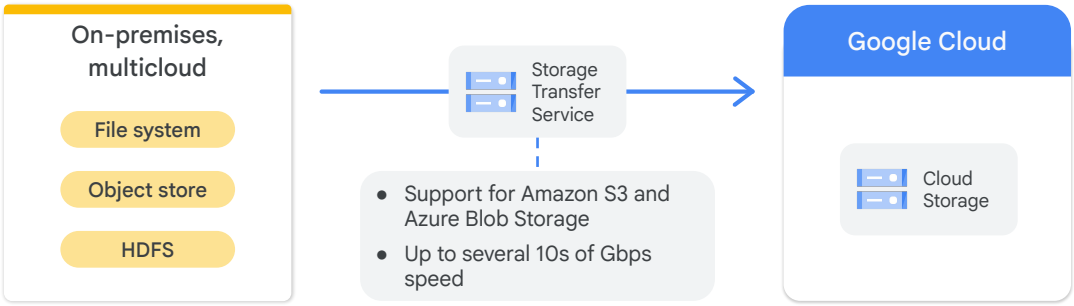
Move large amounts of data with Storage Transfer Service



Medium to large-sized transfers

Scheduling capabilities

Google Cloud

To move larger datasets, consider using Storage Transfer Service.

Storage Transfer Service efficiently moves large datasets from on-premises, multicloud file systems, object stores (including Amazon S3 and Azure Blob Storage), and HDFS into Cloud Storage.

It boasts high transfer speeds (up to tens of Giga bits per second) and supports scheduled transfers for convenient data migration.

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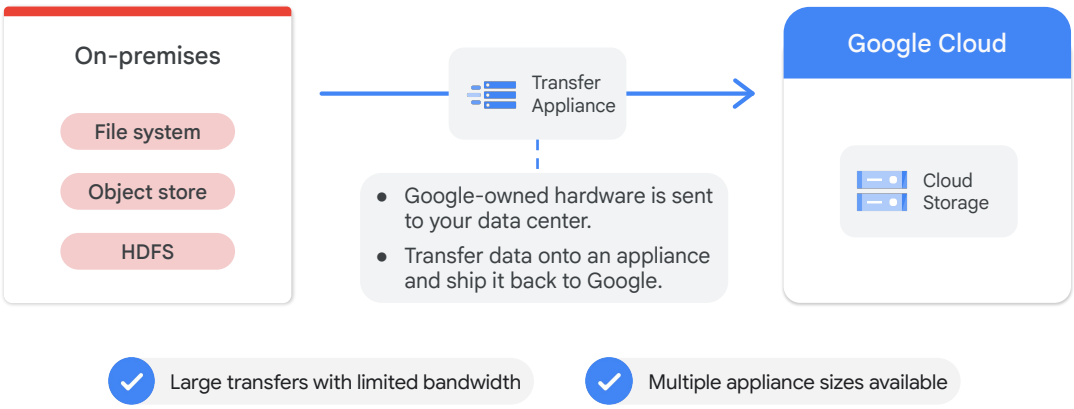
Use Transfer Appliance to move large amounts of data in offline mode



- Large transfers with limited bandwidth
- Multiple appliance sizes available

Google Cloud

Transfer Appliance is Google's solution for moving massive datasets offline.

Google provides the hardware, you transfer your data onto it, then ship it back.

It is ideal for scenarios with limited bandwidth or very large transfers, and it comes in multiple sizes to suit your needs.

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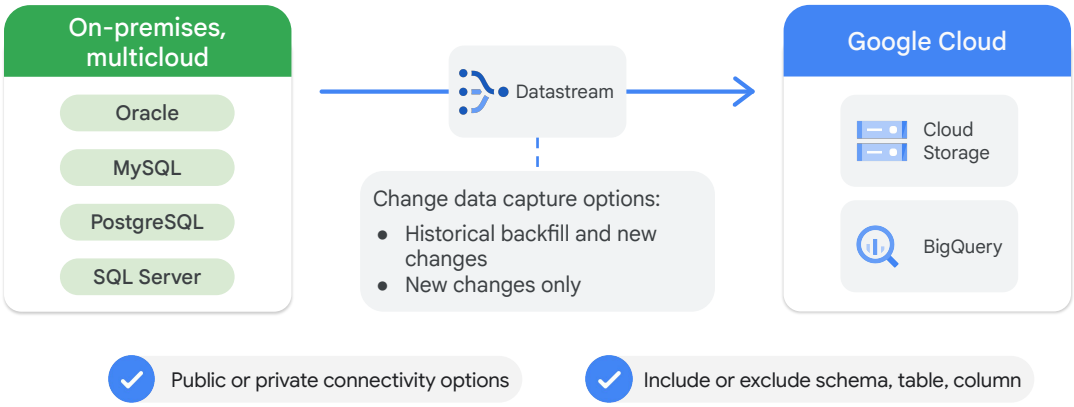
Datastream continuously replicates your RDBMS into Google Cloud for analytics



Public or private connectivity options

Include or exclude schema, table, column

Google Cloud

Datastream enables continuous replication of your on-premises or multicloud relational databases such as Oracle, MySQL, PostgreSQL, or SQL Server into Google Cloud.

Datastream offers change data capture options for historical backfill or allows you to just propagate new changes, with data landing in Cloud Storage or BigQuery for analytics.

You have flexibility in connectivity options and can selectively replicate data at the schema, table, or column level.

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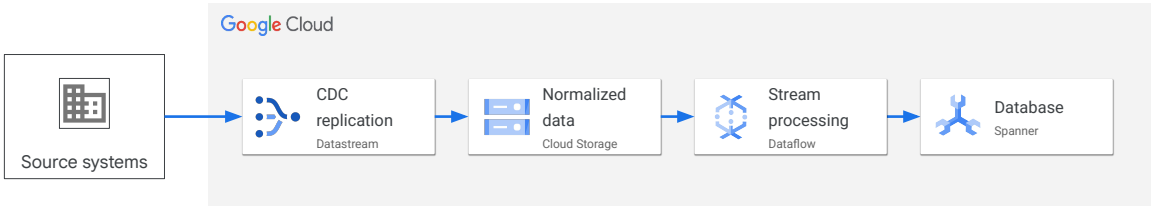
Datastream use cases

- Analytics with database replication into BigQuery
- Analytics with custom data processing in Dataflow into BigQuery
- Data processing using an event-driven architecture
- **Example:** database replication and migration using Dataflow templates



Google Cloud

Datastream enables real-time data replication from source systems for various use cases.

It supports direct replication into BigQuery for analytics, allows custom data processing in Dataflow before loading into BigQuery, and facilitates event-driven architectures.

Additionally, Datastream can be used with Dataflow templates for seamless database replication and migration tasks, making it a versatile tool for integrating data into Google Cloud.

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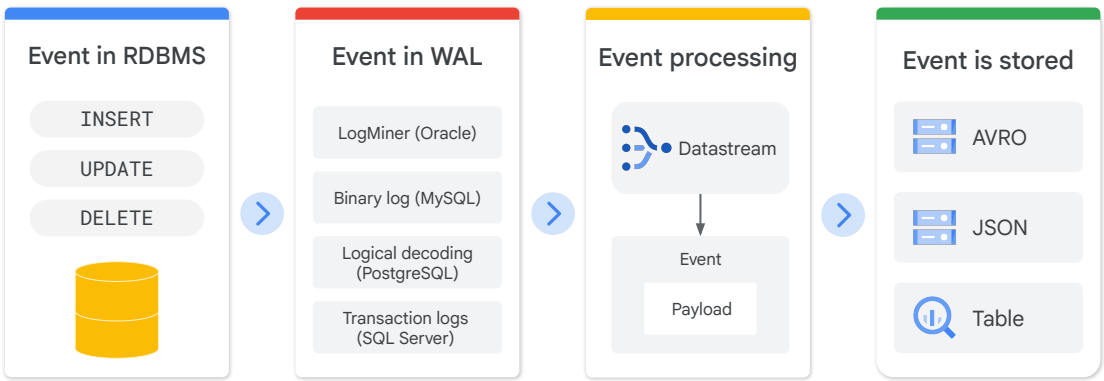
Datastream uses the database's write-ahead log (WAL) to process change events



Google Cloud

Datastream taps into the source database's write-ahead log (WAL) to capture and process changes for propagation downstream.

Datastream supports reading the logging mechanisms for the specific source database such as LogMiner for Oracle, binary log for MySQL, PostgreSQL's logical decoding, and transaction logs from SQL Server.

These change events such as inserts, updates, and deletes are then processed by Datastream and transformed into structured formats like AVRO or JSON, ready for storage in Google Cloud, typically in BigQuery tables, enabling near real-time data replication for analytics and other use cases.

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Event messages contain generic metadata and payload

```
{
 "stream_name": "[...]",
 "read_method": "oracle-cdc-logminer",
 "object": "SAMPLE.TBL",
 "uuid":
"d7989206-380f-0e81-8056-240501101100",
 "read_timestamp":
"2023-11-07T07:37:16.808Z",
 "source_timestamp": "2023-11-07T02:15:39",
 "payload": {
   "USER_ID": "1231535353",
   "FIRST_NAME": "Jane",
   "LAST_NAME": "Smith",
 }
```

generic

- `object`: name of the table or object in the source
- `read_timestamp`: when Datastream read the record
- `source_timestamp`: when the record changed on the source

payload

- `key-value` pairs for each column and the corresponding value

Google Cloud

Datastream event messages contain two main sections: generic metadata and payload.

Metadata provides context about the data, like source table, timestamps, and related information.

Payload contains the actual data changes in a key-value format, reflecting column names and their corresponding values.

This structure allows for efficient and organized data replication and tracking of changes.

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Event messages also contain source-specific metadata

```
"source_metadata": {
   "log_file": "[...]"
   "scn": 15869116216871,
   "row_id": "AAAPwRAALAAMzMBABD",
   "is_deleted": false,
   "database": "DB1",
   "schema": "ROOT",
   "table": "SAMPLE"
   "change_type": "INSERT",
   "tx_id":
   "rs_id": "0x0073c9.000a4e4c.01d0",
   "ssn": 67,
 },
}
```

Oracle example

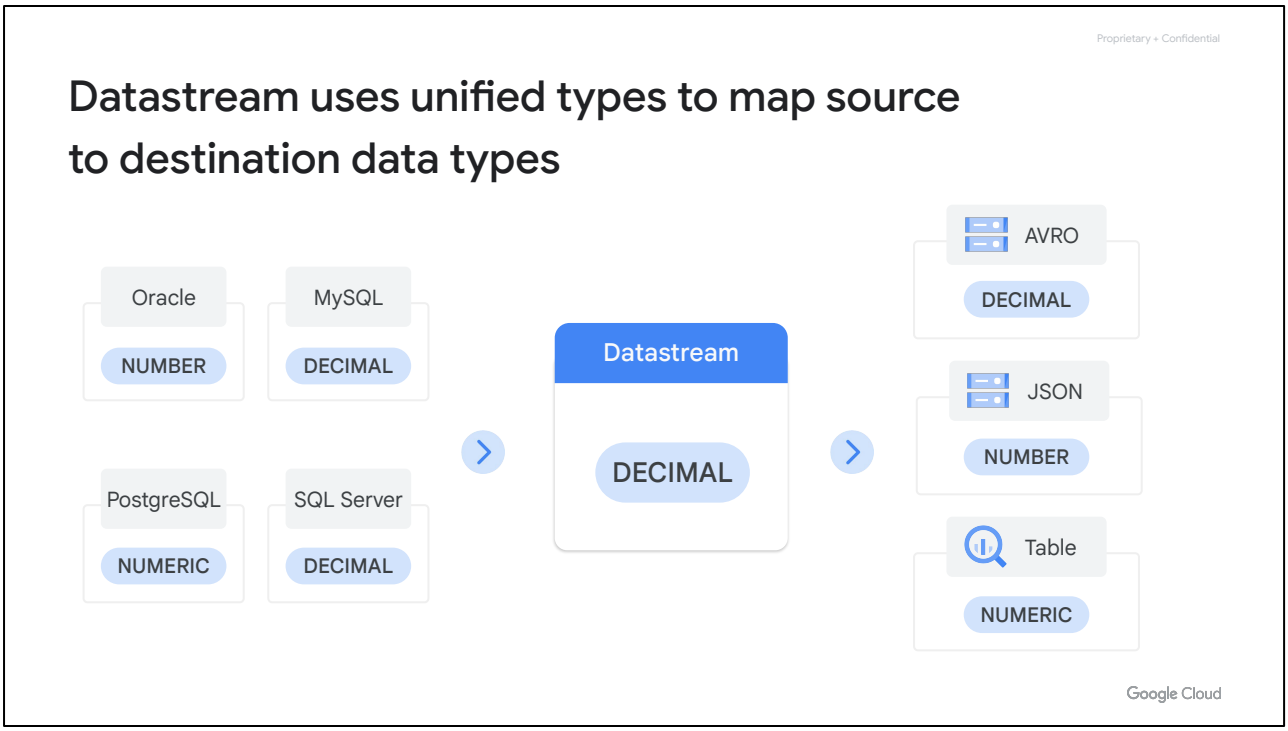
- `database`: the database associated with the event
- `schema`: the schema associated with the table from the event
- `table`: the table associated with the event
- `change_type`: the DML operation (e.g. `INSERT`)

Google Cloud

Datastream event messages also include source-specific metadata in addition to generic metadata and payload.

This metadata provides context about the data's origin within the source system, including details like the database name, schema, table, change type (such as INSERT), and other system-specific identifiers.

This additional information helps track data lineage and understand the context of changes replicated from the source database.



Datastream simplifies data replication by using unified data types to map between different source and destination databases.

This means that regardless of whether your source data is in Oracle as number, MySQL as decimal, PostgreSQL as numeric, or SQL Server as decimal, Datastream will consistently represent it as decimal during replication.

When this data lands in Google Cloud, it can be further transformed into format-specific data types in different file types or destinations, such as Avro as decimal, JSON as number, or stored natively in BigQuery tables as numeric.

This ensures data type consistency and compatibility across different database systems, streamlining the data replication process.

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Compare migration and replication options

	gcloud storage	Storage Transfer Service	Transfer Appliance	Datastream
Transfer type	online	online	offline	online
Source locations	On-premises, Google Cloud	On-premises, Google Cloud, multicloud	On-premises	On-premises, Google Cloud, multicloud
Data range	less than 1 TB recommended	more than 1 TB recommended	7 TB, 40 TB, 300 TB	10,000 tables per stream
Velocity	Batch	Batch (hourly at minimum)	Batch	Batch and Stream
Data format	any	any	any	structured

Google Cloud

In summary, Google Cloud offers several data migration and replication options. The “gcloud storage” command is suitable for smaller, online transfers. Storage Transfer Service handles larger online transfers efficiently. Transfer Appliance is ideal for massive offline data migrations. And Datastream provides continuous, online replication of structured data, supporting both batch and streaming velocities.

Choose the option that best fits your data size, transfer type, and data availability requirements.

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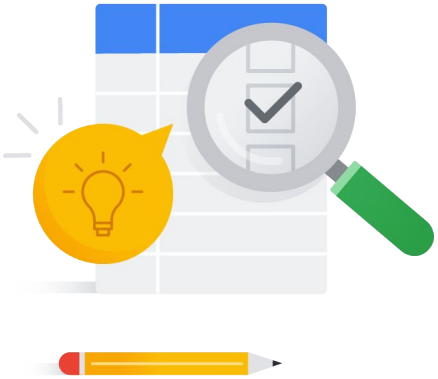
Lab: Datastream: PostgreSQL Replication to BigQuery

45 min

Learning objectives

- Prepare a Cloud SQL for PostgreSQL instance using the Google Cloud console.
- Import data into the Cloud SQL instance.
- Create a Datastream connection profile for the PostgreSQL database.
- Create a Datastream connection profile for the BigQuery destination.
- Create a Datastream stream and start replication.
- Validate that the existing data and changes are replicated correctly into BigQuery.



Google Cloud

In this lab, you use Datastream to replicate data from PostgreSQL to BigQuery.

You prepare and load a Cloud SQL for PostgreSQL instance.

You create Datastream connection profiles for the source and target.

You then create a Datastream processing stream and start replication.

Finally, you validate the replication in BigQuery.